Very High Spatial Resolution Soil Moisture Observation of Heterogeneous Subarctic Catchment Using Nonlocal Averaging and Multitemporal SAR Data

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***Abstract*— A soil moisture estimation method was developed for Sentinel-1 synthetic aperture radar (SAR) ground range detected high resolution (GRDH) data to analyze moisture conditions in a gently undulating and heterogeneous subarctic area containing forests, wetlands, and open orographic tundra. In order to preserve the original 10-m pixel spacing, PIMSAR (pixel-based multitemporal nonlocal averaging) nonlocal mean filtering was applied. It was guided by multitemporal statistics of SAR images in the area. The gradient boosted trees (GBT) machine learning method was used for the soil moisture algorithm development. Discrete and continuous *in situ* soil moisture values were used for training and validation of the algorithm. For surface soil moisture, the root mean square error (RMSE) of the method was 6.5% and 8.8% for morning and evening images, respectively. The corresponding maximum errors were 34.1% and 33.8%. The pixelwise sensitivity to the training set and method choice was estimated as the variance of the soil moisture values derived using the algorithms for the three best methods with respect to the criteria: the smallest maximum error, the smallest RMSE value, and the highest coefficient of determination (R^2) value. It was, on average, 6.3% with a standard deviation of 5.7%. Our approach successfully produced instantaneous high-resolution soil moisture estimates on daily basis for the subarctic landscape and can further be applied to various hydrological, biogeochemical, and management purposes.**

***Index Terms*— Machine learning, nonlocal mean filtering, soil moisture, synthetic aperture radar (SAR), temporal classification.**

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I. Introduction

SOIL moisture retrieval has been one of the main topics of synthetic aperture radar (SAR) research for decades [1]–[3]. The emphasis has long been on bare soil [4]–[6] and agricultural land in general [7], but, recently, uncultivated land types also have been studied [8]–[10]. However, SAR data are less applied and tested in forested areas and in mosaic landscapes of subarctic regions with forest, wetlands, and open tundra. The small penetration depth of the SAR signal in the C-band in high vegetation has been a limiting factor for soil moisture estimation in forests [11]. A high correlation between backscatter and soil moisture at catchment scale but reduced correlation at more detailed scales has been observed in several studies [12], [13], [3]. Current operational global products covering all kinds of land cover types utilizing Sentinel-1 SAR (S1) are provided in 1-km spatial resolution [2], [11], [14], and a new method for soil moisture retrieval in 500-m resolution has been presented recently [15]. The aim is at reaching the spatial resolution of 0.1-km operationally [2], [11].

The soil moisture retrieval applying S1 is currently actively studied in diverse resolutions and ecosystem types [7], [14]–[30]. The methods used have a wide variation. Change detection algorithms are based on the backscattering coefficient dynamical variation between wet and dry

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conditions [9], [14], [15], [17], [19]. Inversion of bare soil and (when needed) vegetation backscattering models is another approach [6], [20], [21], [26]. Polarimetry is also used [22]. However, the retrieval of soil parameters from radar measurements is an ill-posed problem because more than one combination of soil moisture, soil roughness, and vegetation properties can lead to the same electromagnetic response [18]. Hence, the use of diverse machine learning methods is becoming more common in soil moisture retrieval [1], [18], [27]–[30].

The thermal infrared temperature has also been shown to be useful for forested area soil moisture estimation using artificial neural networks (ANN) [24]. Optical image-based normalized difference vegetation index, which is related to leaf area index (LAI) for low vegetation, has been shown to improve the soil moisture estimation accuracy [8], [25], either applying the water cloud model or support vector regression (SVR). In the presence of high vegetation, optical imagery and laser scanner data have been used for analyzing the contribution of vegetation to the backscattered signal [9]. LAI retrieval of the canopy has been shown to be successful using the backscattering ratio of vertical and horizontal polarizations vertically polarized (VV)/horizontally polarized (HH) [31]. Unfortunately, S1 does not provide this ratio. Instead, the backscattering coefficients of VV and cross polarized (VH) polarizations are simultaneously available in S1 data.

The main goal of this study is to obtain spatially and temporally very high resolution (10-m pixel spacing, instantaneous daily) soil moisture maps to scale field-based soil methane (CH_4) flux measurements to a larger area. The area of interest is spatially heterogeneous in terms of land cover and topography. There is a high need to better understand the role of very different surface types in landscape-scale methane budget and especially the role of the upland soils in the CH_4 budget [32]. It is known that, in wet enough conditions, upland soils can act as CH_4 sources, but the interesting question is what kind of spatial and temporal variation in CH_4 exchange can be expected. Within a single catchment, there can be surfaces or patches with high CH_4 emission and those with an order of magnitude lower CH_4 sink, controlled mainly by soil moisture [33]. Hence, normal spatial averaging or Lee filtering [8] to reduce speckle is not applicable. Previously, we have studied midsummer soil moisture patterns and related methane fluxes with a combination of SAR, optical, and topographic data for this region [34]. However, spatiotemporal changes in soil moisture were not assessed. In this article, soil moisture maps including temporal variation are produced for the larger study area, and these maps will later be used in hydrological and dynamic methane models developed for the region.

Nonlocal despeckling can be used to develop high spatial and temporal resolution soil moisture maps. It has been a topic of active research since the first introduction of the principle of nonlocal averaging since 2009 [35]. The main challenge of the method is how to choose the pixels to be averaged nonlocally [36], [37]. One possibility is to leverage information drawn from coregistered optical imagery [38]. This is not always possible when the focus is on analyzing quick changes in the environment, as the area of interest may be cloudy, or the sun elevation is too low. Recently, multitemporal nonlocal despeckling methods have been developed [39], [40], but the aim has been more in analyzing homogeneous targets, such as houses and cultivated fields. In this study, the nonlocal averaging is guided using SAR data only. The new PIMSAR method [41] combines slight spatial averaging (ground range multilook pixels) and temporal averaging with unsupervised classification to derive the guidance for nonlocal averaging.

In this study, the reduced simple ratio (RSR) [42], [43] derived from Sentinel-2 (S2) is used as teaching material for machine learning methods to be developed for VV and VH backscattering for canopy effective LAI estimation. Further on, machine learning methods are used also for soil moisture retrieval using those S1-based effective LAI estimates and *in situ* soil moisture measurements. The advantage of deriving the soil moisture estimation method to be applicable also without optical data is that soil moisture estimation does not then suffer from cloudiness, cloud shadows when the solar zenith angle is large, or smoke plumes of forest fires.

The novelty of this study is that the soil moisture estimation method developed for S1 data without the need for ancillary optical data is applicable also to forests in a gently undulating area, including wetlands, and that the resulting soil moisture map pixel spacing is as high as 10 m, the spatial resolution being of the order of 20 m. In the future, it might be possible to produce instantaneous soil moisture maps for heterogeneous forested scenery every day in such high spatial resolution, which would open completely new possibilities for environmental and forestry applications for S1 data usage.

II. Data

A. Test Area

The test area is Pallas in northern Finland, the corner co-ordinates being 67.9122 °N, 24.0586 °E and 68.0233 °N, 24.2539 °E (see Fig. 1) [34]. The terrain is partly hilly, the altitude varying in the range 260–610 m above sea level. The diffuse tree line transition zone is at about 475–505 m a.s.l above which tundra starts [44]. The Norway spruce tree line has advanced there during the last 100 years [44]. The long-term (1981–2010) annual temperature and precipitation within the area are 1.0 °C and 521 mm, respectively [45]. In spring, the soil moisture maximum in forest precedes the disappearance of snow by one month, but, in the tree line, it is simultaneous with it [46], [47]. National Land Survey of Finland provided a digital elevation model (DEM) and orthorectified aerial images with a spatial resolution of 2 and 0.5 m, respectively. In addition, a cloud-free Sentinel-2 (S2) optical image was available on July 28, 2019. The nearest weather station was at Kenttärova (67.99 °N, 24.24 °E), where the intensity of precipitation and relative air humidity were available at 10-min intervals.

B. SAR Data

Sentinel-1 SAR interferometric wide (IW) Swath data available in ground range format in the time range of May 15–September 12, 2019, were used for this study, altogether 120 images, the only missing day being July 2. The data are delivered in 10-m pixel spacing, but the spatial resolution is

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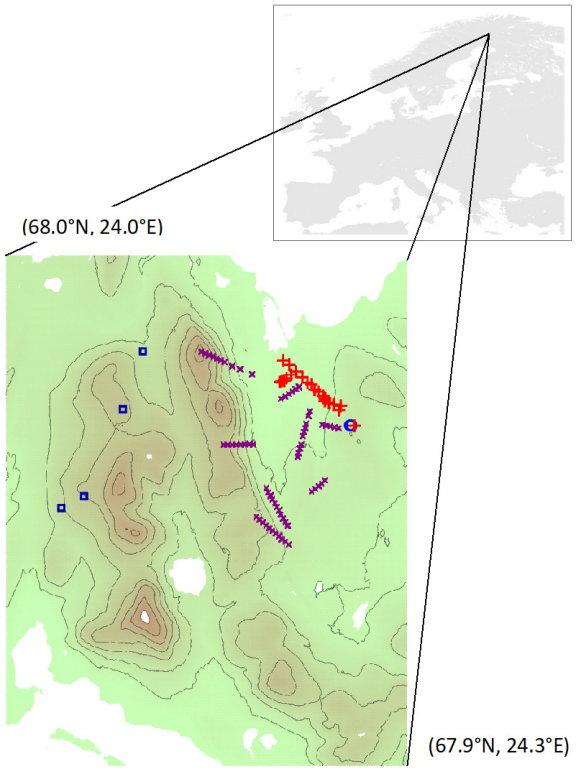


Fig. 1. DEM of the subarctic Pallas test area in northern Finland. The contour interval is 50 m. Open water is masked white. The *in situ* soil moisture measurement points are shown as well: red pluses for the time series points, purple crosses for the intensive campaign points, blue circle for continuous soil moisture measurements at 6-cm depth, and dark blue squares for continuous EMs at 20-cm depth. The area is roughly 12 km × 9 km.

TABLE I
INCIDENCE ANGLE RANGE IN THE STUDIED AREA

Swath	Pass	Minimum incidence angle	Maximum incidence angle
IW1	Morning	33.8°	34.4°
	Evening	33.2°	33.8°
IW2	Morning	39.1°	39.6°
	Evening	38.6°	39.0°
IW3	Morning	43.8°	44.3°
	Evening	43.4°	43.8°

20 m × 22 m. The data were divided into 12 subsets according to polarization, vertical (VV), or cross (VH), swath (IW1, IW2, or IW3), and pass (ascending or descending, i.e., evening or morning, respectively). No discrimination between satellites S1-A and S1-B was made. The incidence angle ranges for the swaths in the test area are given in Table I. The preprocessing steps of all ground range detected high resolution (GRDH) images were: 1) applying precise orbit files; 2) radiometric calibration, nearest neighbor terrain correction applying the DEM; and 3) extracting the subset covering the test area.

C. Tree and Field Layer Data

Biomass and LAI of Norway spruce (Picea abies), Scots pine (Pinus sylvestris) and Downy birch (Betula pubescens), and other deciduous species were available for 131 plots in the test area [34]. The LAI values were calculated using measured tree height and diameter values [48], [49]. The total biomass and LAI values for coniferous and deciduous species and all trees were determined as well (see Table II). In addition, biomass and LAI of the ground and field layer vegetation were determined (see Table III).

TABLE II
FOREST PARAMETER STATISTICS OF 131 PLOTS [34]

Parameter	Species	Median	Mean	Maximum
Biomass [kg/m^2]	Spruce	0.86	2.9	24.0
	Birch	0	1.0	10.1
	Pine	0	1.0	14.7
	Other deciduous tree species	0	0.09	4.8
	Coniferous	1.9	3.9	24.0
	Deciduous	0.03	1.1	10.1
	Total tree	3.4	5.1	24.0
LAI	Spruce	0.88	1.8	11.7
	Birch	0	0.6	6.9
	Pine	0	0.4	4.5
	Other deciduous tree species	0	0.1	2.6
	Coniferous	1.38	2.1	11.7
	Deciduous	0.04	0.7	6.9
	Total tree	2.2	2.8	11.8

TABLE III
UNDERGROWTH VEGETATION STATISTICS OF 131 PLOTS [34]

Parameter	Species	Median	Mean	Maximum
Biomass [g/m^2]	Evergreen shrub	39.7	57.9	212
	Deciduous shrub	61.1	65.3	211
	Forb	1.0	11.3	151
	Graminoid	16.6	23.1	103
	Moss	490	438	610
	Ground and field total	633	595	781
LAI	Evergreen shrub	0.17	0.23	0.81
	Deciduous shrub	0.37	0.40	1.4
	Forb	0.03	0.18	2.4
	Graminoid	0.17	0.27	1.4
	Field layer total	1.0	1.2	3.4

The coefficient of determination for the correlation between canopy biomass and LAI was 0.85, but it was dominated by the smallest LAI values. When only LAI values exceeding unity were taken into account, the R^2 values were only 0.74. The biomass and LAI values of that vegetation did not manifest any correlation.

D. Soil Moisture Data

Discrete volumetric soil moisture values were measured approximately every two weeks in several places at the surface and 10, 20, 30, and 40 cm below the surface using WET-2 and PR2 Profile Probe sensors with an HH2 readout unit (Delta-T Devices Ltd., Cambridge, U.K.,) [34]. In addition, the water level depth was measured biweekly as well. The value of each point was based on three individual measurements within an area having a radius of 5 m so that the three retrievals were typically within one pixel. Hence, the mean values of the triplets were used in the method retrieval. The discrete soil moisture time series was measured during May 28, 2019–September 11, 2019, in 23 points (see Table IV and Figs. 1 and 2). The correlation between the surface soil moisture and the soil moisture measured at various depths

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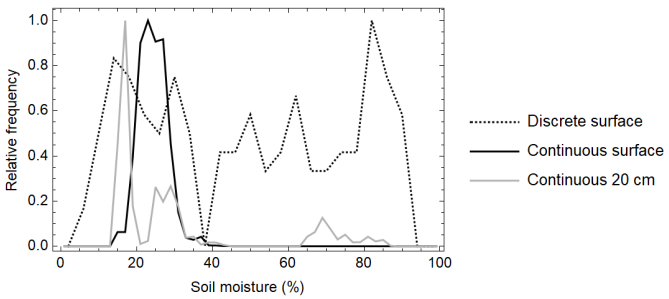


Fig. 2. Relative distributions of the soil moisture measured at the surface at discrete times and continuously and at 20-cm depth continuously.

TABLE IV
DISCRETE *In Situ* SOIL MOISTURE [%] AND WATER LEVEL [cm] STATISTICS. THE NUMBER OF INDIVIDUAL MEASURED VALUES IS *n*

Data set	Parameter	*n*	Minimum	Mean	Maximum
Time series	Surface	678	0.6	49.3	92.3
	10 cm depth	425	0.7	32.5	62.5
	20 cm depth	425	8	46.1	74.4
	30 cm depth	425	9	50.0	84.5
	40 cm depth	424	20.2	54.7	92.8
	Water level	354	0	18.5	71
Intensive campaign	Surface	195	1.3	26.7	91.6

TABLE V
CONTINUOUS SURFACE (6-CM DEPTH) SOIL MOISTURE [%] STATISTICS. THE NUMBER OF INDIVIDUAL MEASURED VALUES IS *n*

Point	Latitude	Longitude	*n*	Min	Mean	Max
S3	67.98703°N	24.24193°E	7838	19	25.3	32
S13	67.98707°N	24.24238°E	7869	15	25.4	56
S17	67.98704°N	24.24193°E	7777	19	21.9	44
S19	67.98705°N	24.24238°E	7880	19	23.0	37

decreased with increasing depth, the coefficients of determination (R^2) being 0.61 (at 10 cm), 0.40 (at 20 cm), 0.32 (at 30 cm), and 0.26 (at 40 cm). The surface soil moisture was negatively correlated with the water level depth, the R^2 value being 0.54. Besides the time series, surface soil moisture was measured in 64 additional points with the WET-2 sensor, Delta-T Devices Ltd., during the intensive campaign carried out on July 3, 2019–July 13, 2019 [34]. Hence, the intensive campaign provides more information about the spatial variation of the surface soil moisture and the times series about the temporal variation.

Continuous soil moisture measurement data at 6-cm depth using soil scout sensors (Soil Scout Oy) were available in the test area with an interval of 15–19 min (see Table V and Fig. 2). These values were taken to directly represent surface soil moisture.

In addition, continuous soil moisture estimates at depths 20, 40, and 60 cm based on electric measurement (EM) using the CS616 and CS650 Soil moisture probes by Campbell Scientific [50] were available in the test area with an interval of 30 min (see Table VI and Fig. 2). Values up to 100% were used in the machine learning training to cover the whole dynamic range, but only values smaller than 100% were included in the later analysis to remove actual liquid water cases from the soil moisture data. As the continuous measurements do not provide the soil moisture at the surface, a linear regression relationship [$\mathrm{sm}_0 = 10.4 \exp(0.0254\ \mathrm{sm}_{20})$] was derived between measured soil moisture values at the surface sm_0 and at 20-cm depth sm_{20} for the 425 points available simultaneously in the discrete data.

TABLE VI
CONTINUOUS SOIL MOISTURE [%] STATISTICS FROM EM. THE NUMBER OF INDIVIDUAL MEASURED VALUES IS *n*

Point	Latitude	Longitude	*n*	Depth [cm]	Min	Mean	Max
GC4	67.970°N	24.095°E	5808	20	14.1	19.6	42.9
				40	16.3	20.8	36.7
				60	17.6	21.4	35.5
GC7	67.967°N	24.083°E	5808	20	13.1	20.4	55.0
				40	17.5	24.2	12.7
				60	13.4	17.6	31.2
GC8	67.988°N	24.115°E	5808	20	32.8	39.7	53.6
				40	9.7	12.7	22.1
				60	7.2	9.0	14.1
Well	68.001°N	24.124°E	3874	20	71.4	79.1	100
				40	78.7	84.8	100
				60	54.1	56.6	64.1

The relationship is rather vague ($R^2 = 0.34$ and the root mean square error is 8.3%), but it was used to convert the EM-based soil moisture dataset from 20-cm depth to the surface, in order to get more extensive spatial coverage and wider land cover type range for the continuous surface soil moisture dataset to be used for deriving the estimation algorithm.

E. Sensitivity of Backscattering to Soil Moisture and LAI_{eff}

In principle, the backscattering coefficient increases with increasing soil moisture and $\mathrm{LAI}_{\mathrm{eff}}$. However, in the hilly terrain of the study area, the backscattered intensity is strongly related also to the local incidence and azimuth angles. Thus, the relationship between the soil moisture, $\mathrm{LAI}_{\mathrm{eff}}$, and backscattering coefficient is not so simple (see Fig. 3). Although the high soil moisture values are common at the high end of a certain $\mathrm{LAI}_{\mathrm{eff}}$ value, opposite behavior appears. Hence, in an area like this, a simple algorithm for soil moisture retrieval is not to be found.

III. METHODS

To preserve the high spatial resolution of the GRDH SAR images and cope with the speckle, a recently developed nonlocal averaging method PIMSAR [41] was applied to each image. For that purpose, the images were first divided into 30 classes based on multitemporal mean and standard deviation values. In each class, the backscattering neighbor locations were sought by sorting the multitemporal mean values of the class. Indices of 49 pixels closest to each other in the multitemporal mean were stored for each pixel in a guidance matrix per pass and swath. The same guidance matrix

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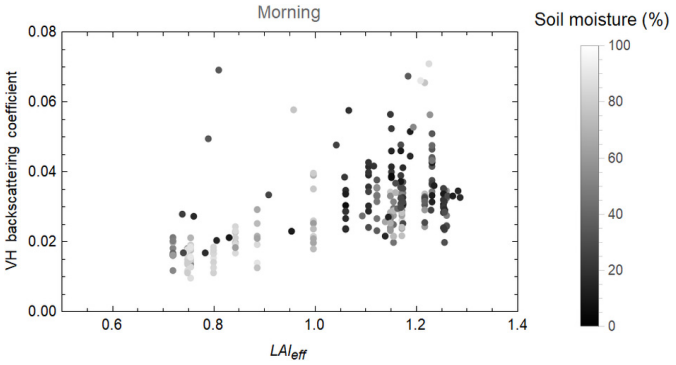


Fig. 3. VH backscattering coefficient versus LAI_{eff} for the morning images and the discrete soil moisture measurement points. The soil moisture is indicated by the darkness of the points.

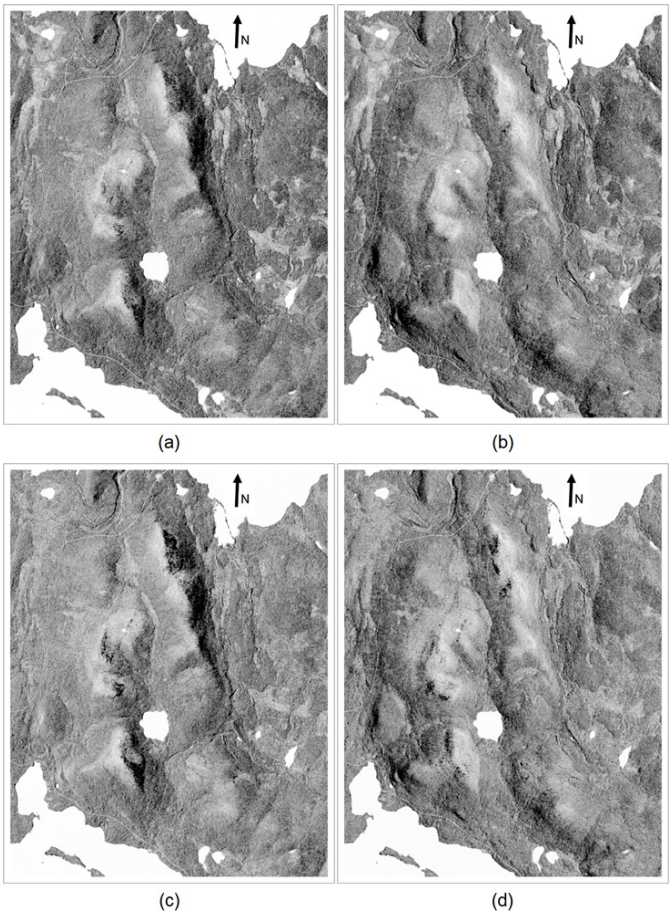


Fig. 4. Nonlocally averaged VH backscattering coefficient of the S1 GRDH images of June 21 (a) and June 22 (b), 2019. The swath is IW3, and the passes are descending (morning) and ascending (evening), respectively. The magnitude increases from white to black to show forests dark and open areas bright. Open water is masked white. (c) and (d) Corresponding images for VV polarization. The area is roughly 12 km × 9 km.

was used for all images of the same pass and swath to derive the nonlocally filtered images. PIMSAR preserves the image mean value with a high accuracy (better than 0.02% for each image). Details of the PIMSAR method are described in [41].

Examples of nonlocally averaged morning and evening VH and VV images are shown in Fig. 4. The hilly topography causes obvious shading in different parts in the morning and evening images, so the soil moisture algorithm has to be derived for morning and evening images separately.

Examples of subareas of original and PIMSAR filtered images are shown in Figs. 5 and 6 for the highest and lowest mean image intensity cases for VH and VV polarizations. Notably, the highest intensity occurs on the same day (September 12), whereas the lowest intensity appears in the VH images on August 7 and in the VV images on May 15.

PIMSAR is able to produce a high spatial structure, which is completely blurred in the original images and their spatial averages. Even a narrow forest road (roughly 3-m wide without nourishing) is visible although the pixel spacing is 10 m.

The relationship between backscattering, leaf area index, and soil moisture is briefly described in Section III-A using a water cloud with gaps approach. The effective LAI (LAI_{eff}) map based on the SAR images was derived using the random forest [51], [52] machine learning method trained with the LAI_{eff} map obtained using the RSR index derived from the S2 image (see Section III-B). Then, the relationship between soil moisture and backscattering was obtained using machine learning methods (see Section III-C) trained with *in situ* soil moisture measurements (see Fig. 7).

A. Ground Contribution to Backscattering

The total backscattering of the vegetated area σ^0 can be described with a water cloud with gaps [7], [53]–[55]

$$\sigma^0 = (1-\eta)\sigma^0_{gr} + \eta\tau^2_{\text{can}}\sigma^0_{\text{gr}} + \eta\sigma^0_{\text{can}}\left(1-\tau^2_{\text{can}}\right) \qquad (1)$$

where η denotes the canopy cover, and σ^0_{gr} and σ^0_{can} are the backscattering contributions of the ground and canopy, respectively. The transmissivity of the canopy τ_{can} is related to the number of scatterers n per unit area and, each having the area A and thickness h, so that the volume of a single scatterer is $V = A\,h$. The amount of water of one scatterer is denoted by W_v (kg/m^3). The total amount of water per unit area in the canopy, i.e., the canopy water content m_v (kg/m^2), is then $n\,W_v\,A\,h$. Typically, the canopy water content is taken to be related to the stem volume, especially when using longer wavelengths. In the C-band, however, also leaves contribute to scattering markedly, as their dimensions are comparable to the wavelength [58]–[60]. The branches of the dominating species of the area, the northern Norway spruce ("candle spruce"), are so small and hidden behind the shoots that their contribution to the C-band is not significant [61]. The question of whether the leaves (needles) or trunks dominate can be checked from the SAR data.

If the trunk-ground double bounce mechanisms were the dominating scattering mechanism, the topography would reduce it so that the slopes facing the radar would produce weaker backscattering than flat areas since the double bounce would not be directed to the radar due to the sloped terrain (see Fig. 8). However, the backscattering images do not show this kind of behavior. Hence, the trunk-ground double bounce backscattering is not the dominant scattering component in the area studied, and the leaves (needles) appear to be the dominating scattering component from the canopy. Consequently, it is reasonable to characterize the canopy with LAI, not biomass.

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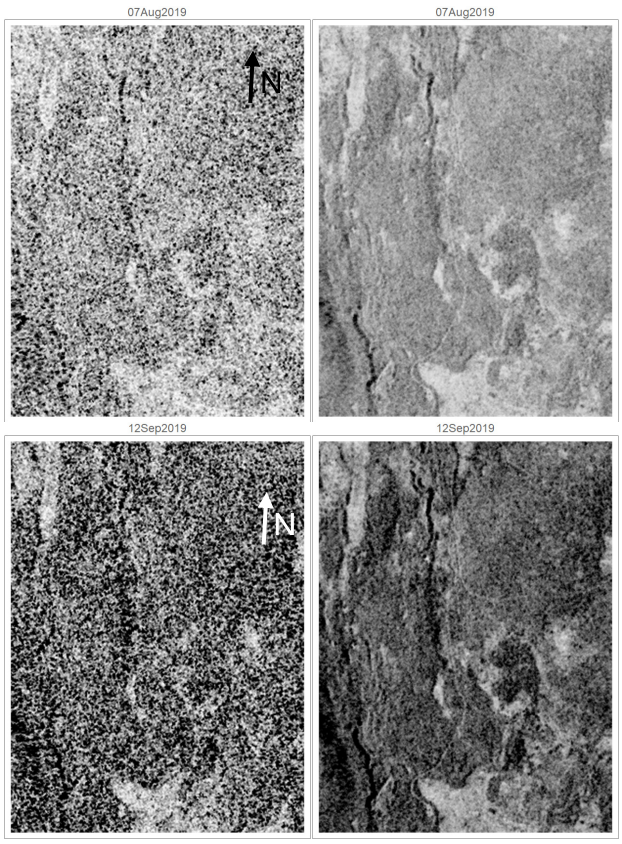


Fig. 5. Subarea of the S1 VH image for the lowest (top) and highest (bottom) image mean backscattering values. The original images are to the left and the nonlocally filtered images to the right. The area covered is 4 km × 3 km.

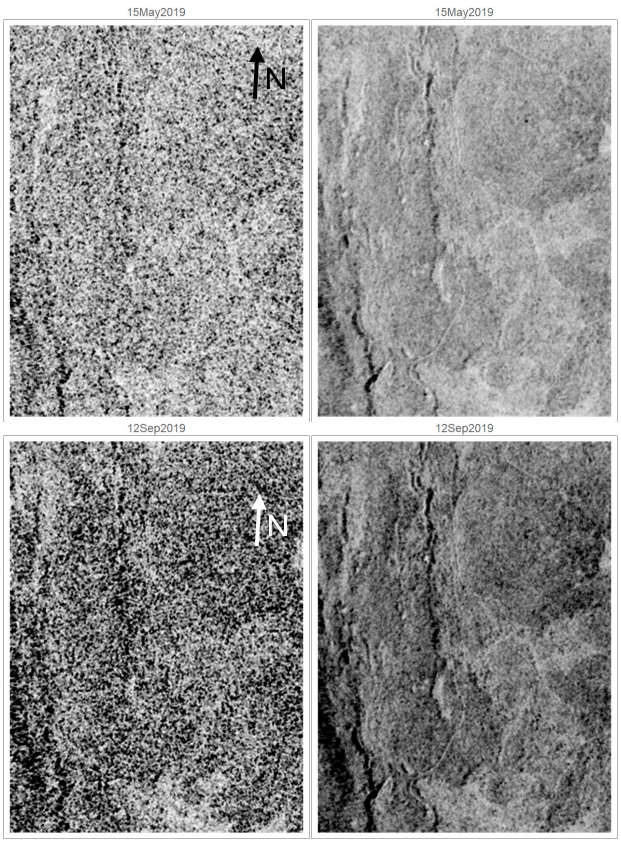


Fig. 6. Subarea of the S1 VV image for the lowest (top) and highest (bottom) image mean backscattering values. The original images are to the left and the nonlocally filtered images to the right. The area covered is 4 km × 3 km.

The leaf area index LAI equals $n \cdot A$. Hence, the total amount of water in the canopy layer can be described by $m_v = W_v \cdot \mathrm{LAI} \cdot h$. The canopy transmissivity is then exponentially decaying with the amount of water per unit area and can be given in the form

$$\tau_{\mathrm{can}} = \exp\left(-B m_v / \cos\theta\right) = \exp\left(-B W_v \mathrm{LAI} h / \cos\theta\right) \quad (2)$$

where θ is the incidence angle of the microwave radiation and B is a coefficient related to the canopy structure. For coniferous canopies, the true hemisurface LAI is about 1.5 times the effective LAI [56], [57]. One should also notice that both optical and microwave remote sensing *LAI* estimates may saturate at large LAI values because the transmissivity is then negligible. As the leaves are taken to have identical thickness, the above equation can be simplified to the form

$$\tau_{\mathrm{can}} = \exp\left(-B' W_v \mathrm{LAI}_{\mathrm{eff}} / \cos\theta\right). \quad (3)$$

Analogically, the backscattering coefficient of the canopy is

$$\sigma^0_{\mathrm{can}} = A m_v \cos\theta = A' W_v \mathrm{LAI}_{\mathrm{eff}} \cos\theta \quad (4)$$

where A' is a parameter related to the structure of the canopy. Because many individual SAR images taken in varying moisture conditions of the vegetation are used, the parameter W_v is explicitly shown both in the description of the backscattering and transmission of the vegetation.

Although $\mathrm{LAI}_{\mathrm{eff}}$ can be estimated and θ is known for each pixel and η, A, and B can be assumed to be constant during July 15–August 31, there are still too many unknowns per pixel to derive the soil backscattering coefficient in a deterministic way. However, this analysis gives motivation for providing the machine learning method with $\mathrm{LAI}_{\mathrm{eff}}$. In addition, the low biomass (see Table II) [59] and sparse structure of the subarctic forests studied [62] support the view that soil moisture retrieval is possible using S1 data.

B. Effective LAI Retrieval

For boreal forest, the RSR index suits for the canopy $\mathrm{LAI}_{\mathrm{eff}}$ estimation using the S2 image [42]. We use the relationship $\mathrm{LAI}_{\mathrm{eff}} = 0.25 \cdot \mathrm{RSR} + 0.27$ [43].

The SAR-based $\mathrm{LAI}_{\mathrm{eff}}$ estimation method was derived for the images June 26–August 7 because the LAI can be assumed to be about constant. As HH polarization was not available in the S1 data, a previously developed LAI retrieval method based on the VV/HH polarization ratio sensitivity to shoot orientation could not be applied [31]. On the other hand, the temporally averaged VH and VV polarization backscattering coefficients of single swaths showed only a weak correlation with the S2 image-based $\mathrm{LAI}_{\mathrm{eff}}$ and saturated at about $\mathrm{LAI}_{\mathrm{eff}} = 1.3$. To benefit from the availability of data from all three swaths, it was decided to derive a machine learning method for $\mathrm{LAI}_{\mathrm{eff}}$ estimation using S1 data.

After removing open water areas and a cloud containing a square from the S2-based $\mathrm{LAI}_{\mathrm{eff}}$ image, there were

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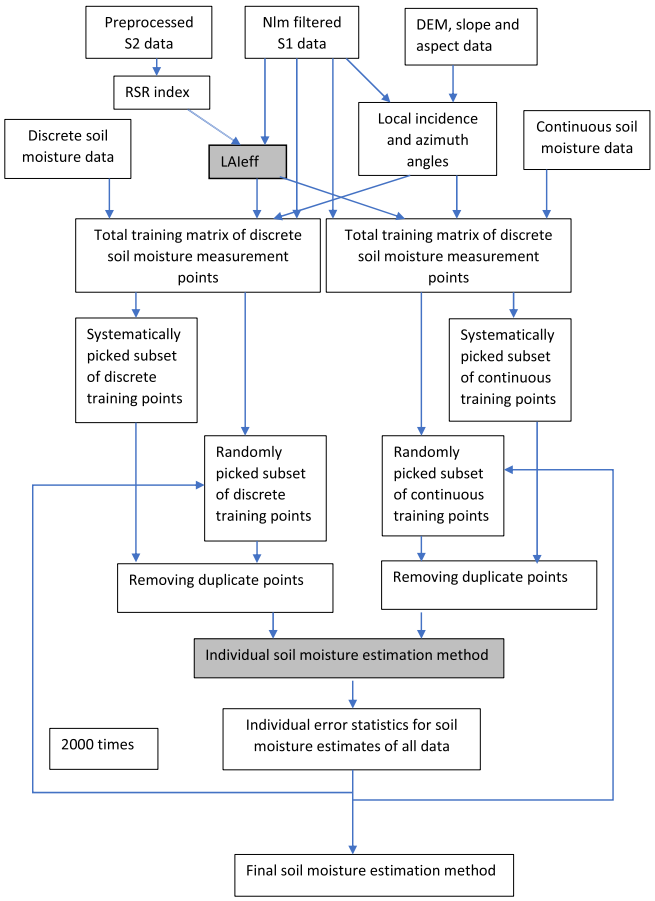


Fig. 7. Flowchart of the soil moisture method retrieval. The machine learning steps are indicated with gray shade. The random picking of the discrete and continuous training points is carried out 2000 times. The number of discretely and continuously measured points is roughly the same.

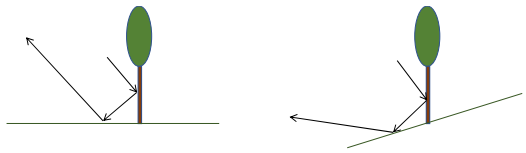


Fig. 8. Schematic of reduced trunk-ground double bounce backscattering of sloped terrain (right) compared to flat terrain (left).

996 897 pixels for machine learning training and validation. The LAI_{eff} distribution showed that a vast majority of the values were in the range of 0.99–1.35, which covers the percentiles from 0.2 to 0.95. In order to avoid their dominance in training the LAI_{eff} method, the train set was chosen so that all values below 0.99 (20%) and above 1.35 (5%) were included in it. For every increment of 0.02 between the limit values 0.99 and 1.35, every 1600th value in the order of magnitude was included. The total number of pixels in the train set was then 484 799. All of the smallest 20% and largest 5% LAI_{eff} values were taken in the training set to optimize the algorithm for the area in question. If a more general algorithm had been the objective, one would have left half of the smallest and largest values in the validation set.

The predictors used for the SAR-based LAI_{eff} algorithm were the nonlocally averaged backscattering coefficients of each swath and pass normalized with the respective average value of the land area in the same image, i.e., 12 individual values per one LAI_{eff} pixel. The normalization was adopted to avoid the effect of varying moisture conditions. No topography-related parameters were used as predictors because the effect of topography is very complex for high vegetation.

The training was first tested using several methods: decision tree, gradient boosted trees (GBT), linear regression, nearest neighbors, neural network, random forest, and Gaussian process. The random forest approach [51], [52] produced the best results with respect to standard deviation; therefore, it was chosen for the LAI_{eff} estimation. The derived algorithm was then applied to the average of individual nonlocally averaged S1 images in the time range June 26–August 7 to produce the LAI_{eff} map for the soil moisture retrieval.

C. Soil Moisture Retrieval

The boreal forest floor is mostly covered with field and ground layer vegetation, including moss and shrub species. In addition, the wetland areas of the test site are also vegetated by shrubs, sedges, and mosses [34]. Hence, the ground backscattering coefficient is a complex combination of soil and low/high vegetation contributions. Therefore, the backscattering models for bare soil [4], [5], [20], [22], [24] were not used for the ground backscattering estimation in this study. Instead, an empirical relationship with the backscattering components and the soil moisture was sought on the basis of *in situ* soil moisture data and machine learning methods.

Due to the field and ground layer vegetation of the boreal forest, the backscattered signal does not trivially depend on the penetration depth in the soil layer [64] but may reveal soil moisture to which the surface vegetation reacts. Hence, it may be possible to relate the radar signal to soil moisture beneath the actual surface level.

A challenge in using the *in situ* soil moisture dataset is that the continuous measurements provide a much larger dataset than the spatially distributed discrete measurements. In addition, soil moisture has a quite marked temporal autocorrelation so that the individual values of continuous measurements are not completely independent. Another complication is that part of the continuous measurements was converted from the values measured at 20-cm depth to the surface soil moisture. Naturally, their somewhat reduced quality will cause some inaccuracy in the surface soil moisture method retrieval.

The following values per pixel were paid attention to in picking the points for the machine learning training set at the time of satellite overpasses: 1) surface soil moisture; 2) day/time of soil moisture measurements; 3) altitude of terrain; 4) local slope angle of terrain θ_l; 5) local aspect angle of terrain φ_l; 6) class (see Section III, beginning); 7) local incidence angle θ; 8) azimuth difference of radar looking direction φ and terrain slope; 9) LAI_{eff}-based on S1 data; 10) $\cos(\theta)$ of S1 data; 11) time difference between the soil moisture measurement and SAR image acquisition; 12) σ^0_{VH};

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13) σ^0_{VV}; 14) $\langle\sigma^0_{VH}\rangle_{land}$; and 15) $\langle\sigma^0_{VV}\rangle_{land}$. Here, $\langle\sigma^0_{VH}\rangle_{land}$ and $\langle\sigma^0_{VV}\rangle_{land}$ denote the average values of σ^0_{VH} and σ^0_{VV} of land pixels of each image.

In principle, one could use all available information (numbers 2–15 of the previous list) to predict the surface soil moisture (number 1). However, then, the machine learning method might find solutions that are actually just typical soil moisture values related to altitude and land cover class or certain day, for example, and the SAR data would provide just noise on top of that. The real test for the suitability of SAR data in soil moisture estimation is to provide the machine learning method no parameters that do not depend on the SAR. Hence, the last nine parameters (numbers 7–15 in the list above) were used as the training information for the predictor to estimate the surface soil moisture (number 1).

Criteria 7 and 8 take into account the longer path the radar signal has to go in the canopy before reaching the forest floor due to the terrain topography. The local incidence angle was provided for the predictor in the form of $\cos(\theta_l–\theta)\cdot\sin(\varphi_l–\varphi)$. The reason for the azimuthal weight is that the signal's penetration in the canopy in hilly terrain depends largely on that. The azimuth difference was provided in the form $\tan(\theta)\cdot\sin(\varphi_l–\varphi)$ to emphasize that the azimuth difference matters more for steep slopes. The radar incidence angle θ was given individually for each pixel of the image, but the radar azimuth angle φ was taken to be the same for all pixels of the pass in question. Criterion 9 (LAI_{eff}) provides the machine learning method information related to attenuation in the canopy. For bare surfaces, taking into account only the local incidence angle of the terrain (criterion 7) would be sufficient, but, for trees, this is not the case. The trunks of trees are vertical also in sloped terrain; hence, the crown scattering is dominated by the radar incidence angle (criterion 10), but the local incidence angle (criterion 7) may have an effect since the tree line to some extent follows the terrain topography. The time between the SAR image acquisition and the *in situ* soil moisture value (criterion 11) will help in taking into account the diurnal variation of the soil moisture.

The main criteria for soil moisture retrieval are numbers 12 and 13 providing the individual backscattering coefficients per pixel in the two polarizations available. The average backscattering values of the images (criteria 14 and 15) are provided in addition to the individual pixel backscattering coefficients because possible winter hardening of coniferous species shows up in changed backscattering level and polarization ratio [62], [64]. Then, the attenuating effect of the canopy is markedly reduced, which affects the soil moisture retrieval. Providing the information about the general status in the area gives a possibility for the machine learning method to decide whether the observed change in single-pixel backscattering is due to soil moisture change only or possibly also to winter hardening.

Although points for the train set were chosen so that they also cover the dynamic range of the timing of the soil moisture measurements and the topography related parameters (numbers 2–5 in the list), those parameters were not offered for the predictor in order to avoid producing information that is not observed by the SAR or related to its viewing configuration. Including the class information (number 6 in the list) to the train set would have improved the results slightly, but it was skipped because soil moisture measurements were not provided for all classes.

The training set was constructed so that it covered the dynamic variation range of all previously 15 listed parameters although only nine of them (all SAR dependent) were used as input for the machine learning algorithm. This was done to have good representativity for the training set. The discrete and continuously measured soil moisture values were picked separately, and their fraction in the training set was roughly even, with a slight preference of the discrete points. Roughly half of the discrete points were picked deterministically and the rest randomly. The continuously measured values were picked similarly in a half-random way. This was done as follows.

The data to be used for training (i.e., the abovementioned nine parameter values per each soil moisture measurement at the time of the satellite overpass) were sorted in order of magnitude separately for each criterion, separately for the discrete and continuous measurements. For the discrete data, every 40th point was then picked in the order of magnitude of the criterion in question to be included in the training set in order to cover the dynamic variation of every parameter. In addition, for each criterion, a random point for every set of successive 41 points in the order of magnitude was also included in the training set. Possible redundant points (i.e., points picked on the basis of more than one criterion) were removed from the training set. The number of discrete measurement points in the training set was about 50% of the available points. The continuous dataset was treated similarly, but, in order to avoid its dominance, only every 9000th point was picked systematically and random points from all sets of 9001 successive sorted points. Possible redundant points were removed. The best morning method had 135 discrete and 126 continuously measured soil moisture values, the best evening method 131 and 122, respectively.

The time difference between the S1 image overpass and the soil moisture measurement was restricted to the range ±1 day (i.e., [−24 h–24 h]). Then, all soil moisture values measured at the surface (including the converted ones) were used as training material. In addition, the time difference between the soil moisture measurement time and the S1 satellite overpass was used both as a criterion of picking points for the training set (besides the already mentioned criteria) and as an input parameter for the machine learning method developed.

First, all the machine learning methods already tested for the LAI_{eff} estimation were again tested for soil moisture retrieval for a few train sets. The GBT method produced the best results with respect to standard deviation and was chosen as the method to be used for the final surface soil moisture estimation method retrieval [66]–[70]. This was carried out using 2000 different half-randomly picked training sets. The reason for GBT being the best method for the soil moisture estimation and the random forest method for the LAI_{eff} estimation is that GBT is able to provide better estimates for a finite variation range (0%–100% for soil moisture) than random forest, which

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is otherwise a very good method (fast, stable, and usually robust to outliers).

The R^2 values, RMSE, and maximum errors of the predictor applied to all data were determined separately for all train sets, the whole material, and the validation set, which contained the points not included in the training set. The predictor producing the smallest maximum error for the whole material was chosen to be used. The reason for choosing the min–max norm instead of the highest R^2 or smallest RMSE value was to avoid overfitting to the spatially relatively sparsely representative material. Separate predictors were derived for the morning and evening images due to the hilly topography. The sensitivity of the method to the choice of training points and the chosen method was tested by producing soil moisture maps with three best point sets (with respect to min–max, RMSE, and R^2) and calculating the difference of the soil moisture estimated per pixel.

Machine learning methods may not apply outside the variation range of the parameter values used for training. In addition, the major inaccuracy may appear in areas, which are geographically essentially different from those included in the train set. Therefore, masks for morning and evening images were constructed for areas with possibly unreliable soil moisture estimates. The criteria checked were: 1) altitude (max); 2) terrain slope (max); 3) terrain aspect; 4) class; 5) local radar incidence angle; 6) local radar look angle; and 7) $\mathrm{LAI_{eff}}$ (max). It should be noticed that the terrain aspect and local radar look angle matter only when the slope is relatively steep. The limit value of 0.2 was used for the slope to mask using the aspect and local look angle. Complications were caused by the $\mathrm{LAI_{eff}} = 0$ values at high altitude above the tree line. No soil moisture measurements were available there, yet the area is geographically very different from low altitude areas of low $\mathrm{LAI_{eff}}$ values. Hence, an additional criterion of $\mathrm{LAI_{eff}} = 0$ values at an altitude higher than 450 m a.s.l. was used for masking areas of reduced reliability.

IV. Results

The RMSE of the S1- and S2-based $\mathrm{LAI_{eff}}$ estimates was 0.10 for the train sets and 0.14 for all pixels. The relative 80% variation range of the S2-based $\mathrm{LAI_{eff}}$ estimate was in the whole land area 0.84–1.31, and the corresponding variation range of the S1-based $\mathrm{LAI_{eff}}$ estimate was 0.84–1.25. Hence, the microwave estimate had a slightly smaller dynamic variation than the optical estimate, and its mean value of 1.09 was also slightly smaller than the S2-based mean value of 1.12 (see Fig. 9). However, the spatial variation of the $\mathrm{LAI_{eff}}$ estimates was quite similar (see Fig. 10), the microwave estimate having slightly better pixel spacing (10 m) than the optical (20 m) due to the lower spatial resolution of the shortwave infrared (SWIR) channel of S2.

The soil moisture predictors derived separately for morning and evening images are characterized statistically in Table VII and Fig. 11. The large difference between the maximum error and 90% percentile value suggests that the data contain a small number of points that deviate largely from the general behavior.

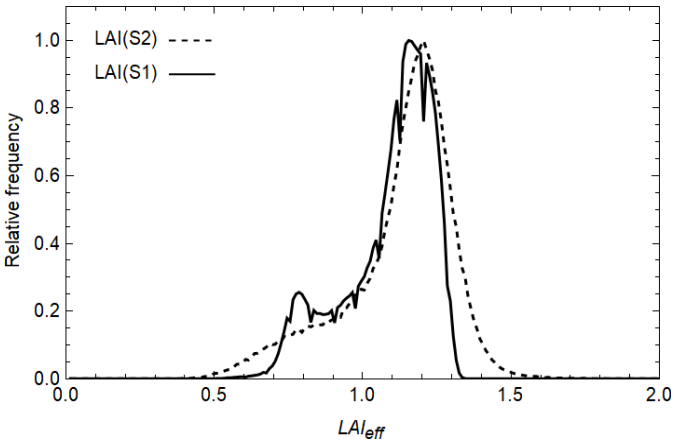


Fig. 9. Relative distribution of the S1- and S2-based $\mathrm{LAI_{eff}}$ estimates.

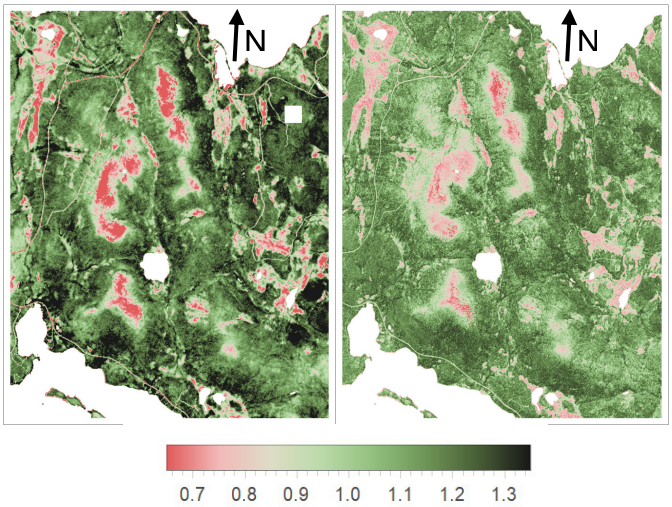


Fig. 10. $\mathrm{LAI_{eff}}$ estimated from the RSR index of the S2 image (left) and using the random forest method applied to the VH and VV backscattering of the S1 images (right). Open water and the cloudy square in the S2 image are masked white.

TABLE VII

Statistics for the Predictors of S1-Based Surface Soil Moisture [%] Estimation Within One Day of S1 Overpass Using Min–Max Norm

Images	Parameter	Trainset	Validation set	All points
Morning	n	260	34836	35096
	offset	4.49	1.33	1.45
	slope	0.90	1.03	1.02
	R^2	0.94	0.87	0.87
	RMSE	5.9	5.7	4.3
	90% error percentile	8.0	9.0	9.0
	Max error	34.1	30.5	34.1
Evening	n	253	34375	34627
	offset	5.1	1.44	1.50
	slope	0.86	0.98	0.98
	R^2	0.91	0.90	0.90
	RMSE	6.8	4.6	3.3
	90% error percentile	10.9	6.8	6.8
	Max error	33.6	33.9	33.9

The relationship between the soil moisture estimation error and the parameters 1–14 listed in Section III-C revealed that there is a slight systematic dependence of the error on the Julian day of the soil moisture measurement. The error is more typically negative in May and early June. Since that

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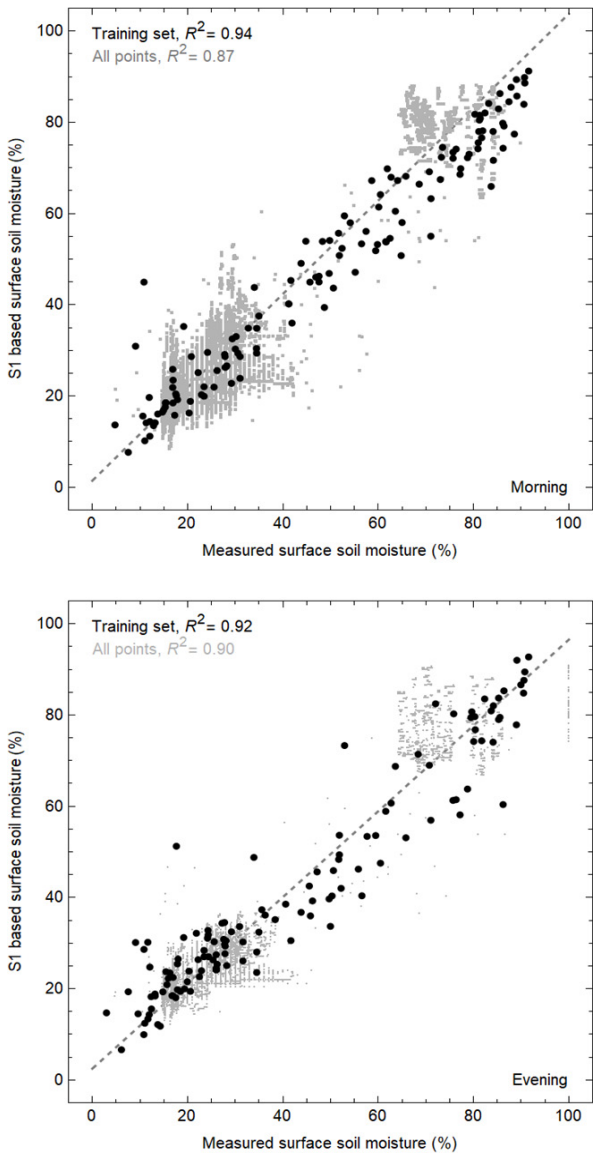


Fig. 11. Estimated surface soil moisture versus the measured values for the train set and all data using the algorithm derived for morning images (top) and evening images (bottom).

period coincides with the start of the actual growth season of the canopy (and field and ground layer vegetation), the real LAI_{eff} is probably lower than the value derived using the summer images (see Section III-B) and used for the whole period. Then, the fraction of backscattering due to the ground would be larger than assumed, which would lead to expected soil moisture being higher than the actual soil moisture and, thus, cause negative estimation error. A first-order polynomial was fit to the error estimate dependence on the Julian day, but the R^2 values were very small, 0.03 for morning and 0.04 for the evening. Hence, this systematic error trend is insignificant compared to the general estimation accuracy.

The estimated seasonal variation of soil moisture of the discrete measurement points was mostly in the range of the measurement variation of the three individual measured values with an accuracy of the RMSE variation of the estimation

TABLE VIII

STATISTICS FOR THE PREDICTORS OF S1-BASED SURFACE SOIL MOISTURE [%] ESTIMATION WITHIN ONE DAY OF S1 OVERPASS FOR THE TIME SERIES OF THE DISCRETE POINTS

Images	Parameter	Discrete points
Morning	n	202
	offset	6.3
	slope	0.84
	R^2	0.88
	RMSE	6.5
	90% error percentile	15.3
	Max error	30.1
Evening	n	165
	offset	8.2
	slope	0.80
	R^2	0.77
	RMSE	8.8
	90% error percentile	24.2
	Max error	32.9

method. The morning images (see Fig. 11 and Table VIII) produced slightly better results than the evening images (RMSE = 6.5 and R^2 = 0.88 for morning and RMSE = 8.8 and R^2 = 0.77 for evening images). Since soil moisture has a strong temporal autocorrelation, except for the beginning of precipitation events, the large continuously measured datasets dominate the statistics of the whole dataset in Table VII, and the RMSE values are too small to be represented in general. Hence, the statistics of Table VIII are considered to better represent the quality of the soil moisture maps.

Although the statistics of the whole dataset containing also the continuous soil moisture measurements look about as promising for the evening dataset as for the morning dataset (see Table VII), the GBT method derived for the morning dataset produced clearly better soil moisture estimates for the time series of the spatially distributed discrete *in situ* points (see Table VIII, Fig. 12, and Supplementary Fig 1). The soil moisture estimates derived using the morning images are mostly within the range of the *in situ* triplet variation measured during the same day, taking into account the RMSE value of the S1-based soil moisture estimates. For evening images (see Table VIII), the agreement of the derived soil moisture estimates with the *in situ* measurements of the same day had more scatter.

The potential of the method for daily soil moisture estimation on a pixel basis was studied using continuous surface soil moisture measurements. Two of the measured points were within one S1 pixel, the other two being situated in two neighbor pixels. The distance between those four points varied in the range of 4–24 m. The estimated soil moisture per day is shown for those three pixels in Fig. 13. It is obvious that the variation within the pixel is high, and it is not known, for example, whether s13 or s19 represents the dominating characteristics of the pixel they are in. Although the diurnal variation of those points is rather small compared to the soil moisture estimation method uncertainty, the estimated values seem to catch the main features of the soil moisture seasonal variation. In addition, the difference in soil moisture values of neighbor

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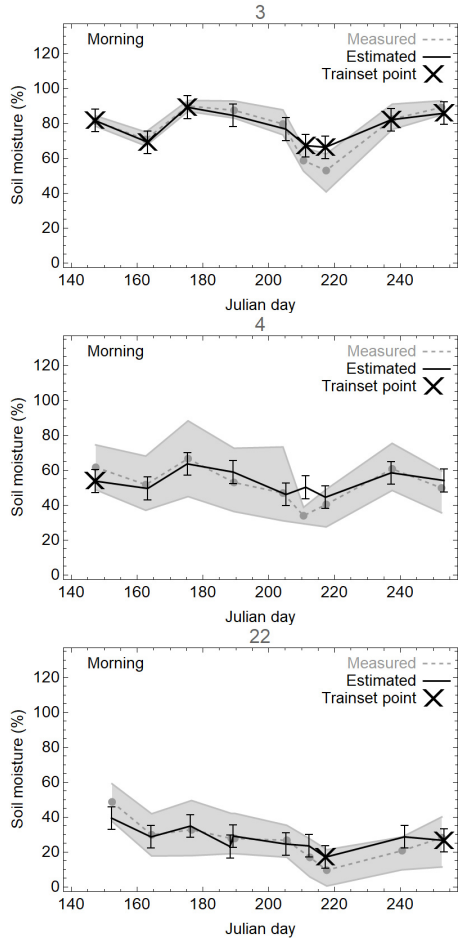


Fig. 12. Examples of the estimated surface soil moisture versus the measured *in situ* values for the discrete points using the algorithm derived for morning images. The gray points show the individual triplet means, and the shaded gray area shows the variation of the triplet values. The black points are the estimated values derived from the morning images of the same day, i.e., a few hours before the *in situ* values. The black error bars are the RMSE values for the GBT method derived for the morning images and applied to the discrete points only (see Table VIII). The points marked with a cross were included in the training set.

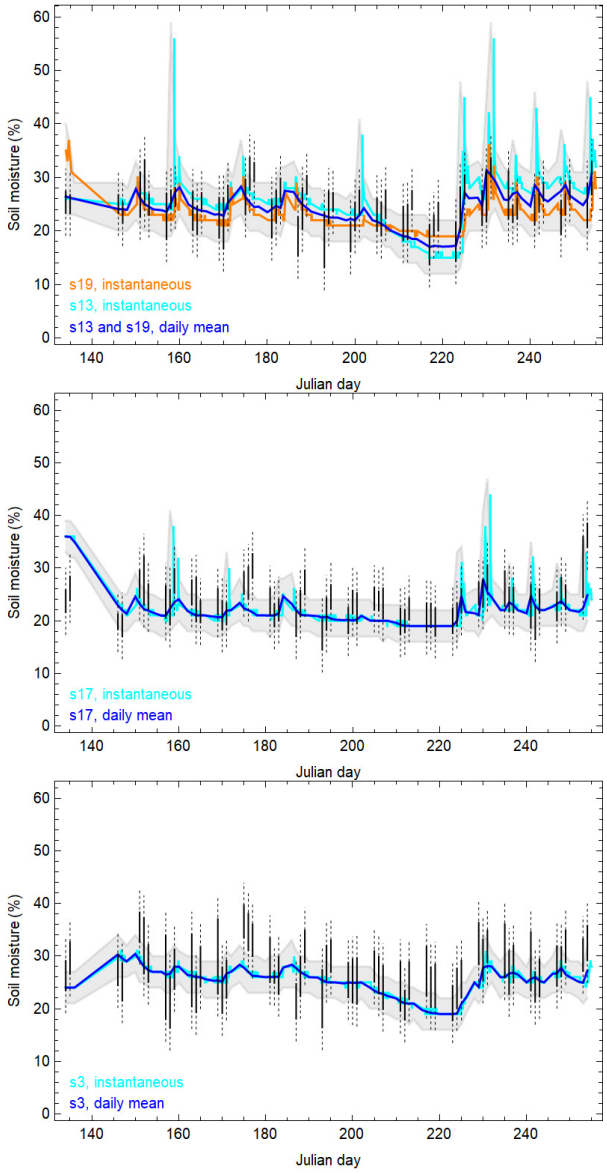


Fig. 13. Daily averaged measured soil moisture for three neighbor S1 pixels (blue) and the variation range (gray band). The individual instantaneous measured values are shown in cyan and orange. The variation range of corresponding estimated soil moisture values based on morning S1 images and the individual *in situ* measurement times are shown as black vertical lines. The variation range is extended by the method overall uncertainty of 4.3% (see Table VII) indicated as dotted lines.

pixels shows also in the estimated values. However, also the difference in swaths (successive points) appears occasionally in the estimated soil moisture curves. Partly, this suggests that the machine learning method is not completely able to hide incidence angle variation-related effects. Partly, this may be a result of not having the same soil moisture variation in the training set per swath, as the precipitation events were not evenly distributed between swaths.

Only three points of the discrete data were located in neighbor pixels during the intensive measurements. A comparison of nearby *in situ* and estimated soil moisture values supported the view that the soil moisture can, in principle, be estimated on a pixel basis, but the number of points (Supplementary Fig. 2) is far too small to prove it.

Altogether 60 instantaneous midday soil moisture maps, one per day, were generated using morning images in order to perceive the temporal variation of the soil moisture of the whole area (see Supplementary Fig. 3). Examples are shown in Fig. 14.

May 21 is about the time of snowmelt at the tree line and, thus, represents an early summer case of high soil moisture. August 12 was preceded by a dry period of two weeks, and the difference to the early summer situation is obvious [see Fig. 14(b)].

One example of the effect of precipitation on soil moisture is shown as a difference between August 13 and August 12 [see Fig. 14(c)]. Autumn soil moisture conditions are clear in the image of September 12 [see Supplementary Fig. 3]; the difference of which to the situation of May 21 is shown in Fig. 14(d). The masks for morning and evening images are shown in Fig. 15.

The sensitivity of the result to the train set and method was studied by comparing the soil moisture maps produced

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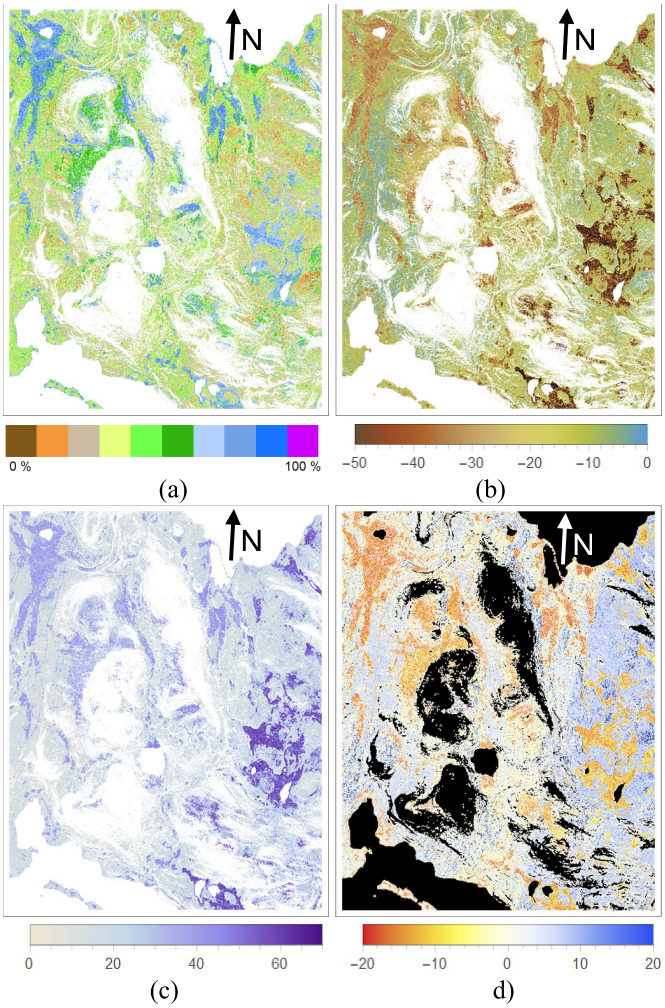


Fig. 14. Calculated soil moisture map for (a) May 21, 2019. Difference maps for (b) August 12–May 21, 2019, (c) August 13–12, 2019, and (d) September 12–May 21, 2019. All maps are calculated for midday (UTC 10) using morning images of S1. Open water and masked areas are white in (a), (b), and (c) and black in (d).

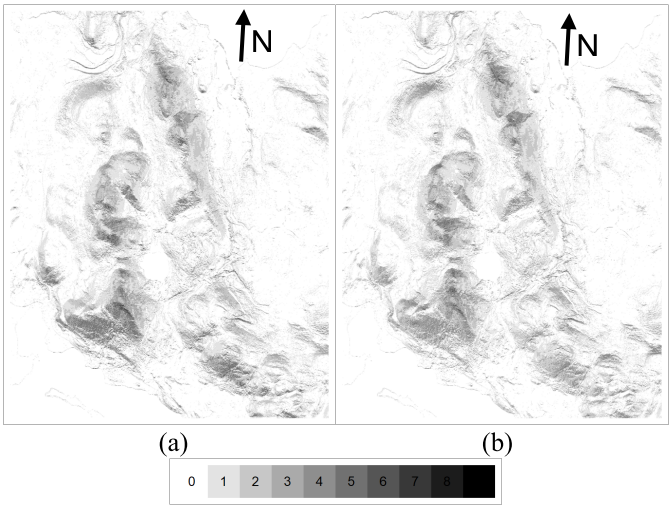


Fig. 15. Mask layers for the soil moisture retrieval using S1 (a) morning images and (b) evening images. The gray level indicates how many criteria the pixel are masked.

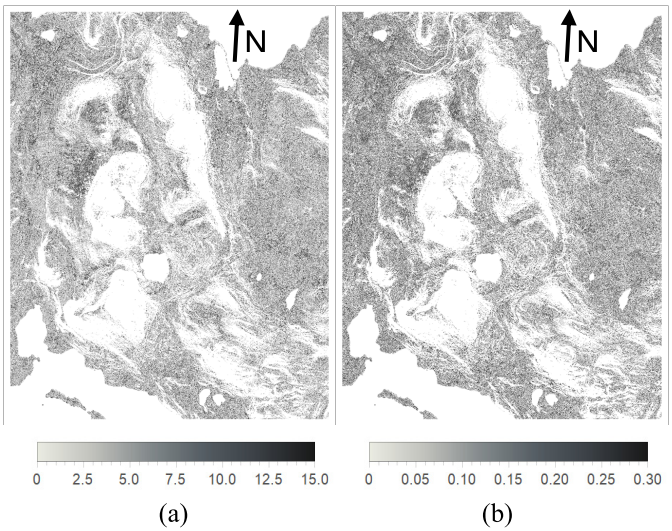


Fig. 16. (a) Masked difference (in soil moisture%) of June 21, 2019, and (b) relative difference between the largest and smallest soil moisture estimate values derived using the best morning algorithms based on: 1) the min–max norm; 2) the smallest RMSE; and 3) the largest R^2 values.

using morning images, the chosen best min–max method, and the methods based on the lowest RMSE or the highest R^2 value. The difference between the maximum and minimum values of the soil moisture estimates produced using the three methods was used to generate a map of pixelwise sensitivity to the chosen train set and method (see Fig. 16). The corresponding relative sensitivity was obtained by dividing the difference of the largest and smallest values by the median value (see Fig. 16). The average difference was 6.3% with a standard deviation of 5.7%, both given in absolute soil moisture values. Those values are quite in line with the quality statistics of Table VIII.

V. Discussion

Our approach successfully produced very high resolution (∼20 m with 10-m pixel spacing) soil moisture estimates on a daily scale to a subarctic landscape. Generally, estimates were matching with 6.5%–8.8% RMSE accuracy (in absolute soil moisture percentages) and R^2 values within 0.8–0.9 in all study periods and were especially good for morning images. The R^2 values obtained using the whole datasets (0.87–0.90) are probably so high and the RMSE values (3.3%–4.3%) so small because of the high temporal autocorrelation of the continuously measured data, as soil moisture changes abruptly only at the beginning of a precipitation event. Hence, we use here the higher RMSE and lower R^2 values obtained for the discrete time series points to characterize the accuracy of the method.

Results agreed with previous SAR studies done in other than agricultural areas, such as the Mediterranean [24], Polish wetland [8], and subarctic conditions [10], in showing that C-band SAR data can be used for soil moisture retrieval in diverse landscapes. Our approach, benefiting from the completely new PIMSAR nonlocal despeckling method [41], is unique and also produced good estimates in forest areas, which have been mostly lacking in previous S1 data using studies [24]. In boreal forests, one complication is the field and

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ground layer vegetation, which may be quite thick, when the canopy is relatively small. However, the field and ground layer vegetation will reflect the surface soil moisture conditions. In addition, the crown shape of northern Norway spruce is very narrow so that a larger fraction of the ground is visible between the trees than in southern Finland for the same LAI value. PIMSAR also enables retrieval of as accurate (∼10%) soil moisture maps without the need for optical data in higher spatial resolution (∼20 m) compared to the previously mentioned three studies: 30 [24], 500 [8], and 100 m [10].

It was a challenge for the method development that the spatially distributed data were not measured at the times of the satellite overpasses, the continuous measurements were available practically only in five locations, and in four of them, the measurements were carried out only at 20 cm depth or deeper. In addition, spatially, the dataset was rather small for machine learning method development. Soil moisture values were available only for less than 0.1% of the land pixels, yet the results are convincing and indicate that it is possible to develop a high-resolution soil moisture retrieval method that uses S1 images as input. Here, we used one S2 image for LAI_{eff} training, but, in an area, where notable changes in the canopy do not take place, one could probably use the same LAI_{eff} map for more than just one summer so that only S1 data would be needed for the soil moisture map processing. This is a major advantage in areas that are often cloudy or at times of poor illumination.

Including LAI_{eff} in the training set provided the machine learning algorithm information about the attenuation of the radar signal in the canopy and prevents the method from finding solutions that are not consistent with that. In the study area, there was a vague increasing trend of soil moisture with decreasing LAI_{eff}. This might favor finding high soil moisture values in the low LAI_{eff} area. However, the trend was so weak ($R^2 = 0.28$) that it should not have a marked effect on the results.

The reason for the morning images to have slightly smaller RMSE values than evening images in this study contrarily to the Polish wetland study [8] is partly due to the topography and location of the measurement points (see Fig. 1) [34]. Several discrete measurement points were in the distal slopes of the evening SAR images and prone to shading. In addition, this study is profited by the continuous measurements, which provided an intelligent way to relate midday measurements to morning and evening S1 images.

Soil moisture estimates are very important for various applications including climate, hydrological, biogeochemical, and management purposes [71], [72]. As subarctic areas are facing very drastic changes due to climate change, including changes in precipitation and snowmelt timing [73], it is essential to have high-resolution estimates from soil moisture conditions in time and space [74]. Our methodology could be further applied to other subarctic and arctic landscapes to provide soil moisture estimates for: 1) ecohydrological and vegetation water usage studies [75]; 2) to improve greenhouse gas estimates from forest lands; 3) to map influences of different land use actions such as logging at the landscape level; and 4) to assess the forest fire risk. In addition, better estimates from soil moisture can further help local land managers to identify spatially important areas from a soil moisture point of view.

Using machine learning methods requires extensive data sets to obtain generally applicable methods. When the data availability is limited in space or time, there is a considerable risk of overfitting. Hence, the results of this study rather demonstrate the capability of using S1 SAR images for seasonal soil moisture variation in one study area than provide a final method to be used in other areas. With larger amounts of soil moisture data, a generally applicable method could be developed, for example, using global soil moisture network databases.

The results obtained using machine learning methods depend largely on the training data provided. Here, the soil moisture distributions of the discrete datasets were slightly bimodal (see Fig. 2), and the continuous datasets were dominated by soil moisture values below 40%. The goal was to achieve a homogenous error structure, i.e., the low, average, and high soil moisture estimation accuracies should be about the same. For this purpose, the training data were chosen to cover the whole variation range of all parameters affecting the soil moisture about evenly. In addition, the choice of the best method was based on min–max norm so that even the rare values are estimated at least at the accuracy of the min–max value. However, it depends on the application what kind of an error structure is desirable. If one would prefer to have the best accuracy for most cases at the expense of highly erroneous rare values, one would choose another norm (for example least squares) and pick the training set so that the most common values are most common also in that.

The effect of the slightly bimodal soil moisture distribution on the error statistics was checked by deriving the RMSE, 90% quantile, and max error values also separately for soil moisture ranges 5%–50% and 60%–90% (see Table VII(A) and (B) in the Supplement material). For the lower range, the soil moisture RMSE was 5.2% and 3.9% for all points in the morning and evening, respectively. Hence, the estimation accuracy was about the same as for the whole material. For the high soil moisture variation range, the corresponding RMSE values were 10.4% and 9.8%, respectively. Hence, it was slightly poorer than for the whole material. Very probably this is mainly due to the inaccuracy of the conversion of the values measured 20-cm depth to surface values (see Fig. 2 and Tables IV–VI). The RMSE value for the conversion formula derived using the discrete measurements was, namely, only 8.3%. The fraction of the converted data was for morning cases: 96% for the high range, 35% for the low range, and 40% for all points. For evening, the corresponding values were 97%, 36%, and 40%.

Balancing between spatially distributed discrete measurement points and a few continuous measurement points was a major issue in this study. In addition, the instruments used for continuous measurements were not identical to those used for the discrete datasets. This inevitably affected the results to some extent. Especially, transforming the soil moisture measured at 20-cm depth to surface values certainly caused extra uncertainty, which is most probably seen in the scatter

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of points in Fig. 11. However, the achieved soil moisture estimation accuracy was of the order of that of the *in situ* measurements.

In the autumn, underestimation of the soil moisture may occur because of reduced canopy scattering due to the fall of deciduous leaves and possibly also the decreasing impedance of Norway spruce needles, which may take place already in August, while the trees prepare for the winter [64]. However, precipitation will mask this effect.

Although topography constitutes a major complication for the use of SAR data, the machine learning method was able to adapt to hilly terrain surprisingly well. Only the completely shaded areas or slopes of serious foreshortening will not be suited for soil moisture estimation. The use of both morning and evening data is complicated by the hilly terrain because the shaded or foreshortened pixels are in different locations. In addition, the discrete *in situ* measurements were taken in the daytime after the morning SAR image retrieval, whereas the evening image overpasses succeeded the discrete *in situ* measurements. Comparing the average backscattered intensities of the images to precipitation and air humidity [see Supplementary Figs 4 and 5] showed that the dynamic variation of the evening image intensity values is smaller than that of the morning images, especially in late summer. It may be that vegetation reacts differently to varying moisture conditions during late summer than at the beginning of summer. Hence, also, for this reason, deriving a consistent soil moisture estimation method functioned better using morning images.

The difference of swaths was taken into account by the local incidence and azimuth angles. Within the dynamic range of the swath incidence angles corresponding to the *in situ* measurement sites, the soil moisture distributions of the whole season were very similar for all swaths. However, it may be that results would improve, if the LAI_{eff} values were derived separately for each swath, as the signal penetration in the canopy depends on the incidence angle. When a larger *in situ* dataset is available, specific methods adapted to the diverse swaths can be derived, which will improve the results even more.

VI. Conclusion

In this study, an S1-based method for assessing soil moisture in a heterogeneous landscape was developed. It was demonstrated that, with machine learning methods, PIMSAR nonlocal mean filtered S1 pixels of 10 m can be used for accurate instantaneous surface soil moisture estimation. The RMSE values were within 6.5% for the morning images and 8.8% for the evening images, which is of the order of the accuracy of the *in situ* measurements. The soil moisture estimates followed consistently the seasonal variation of the *in situ* values. Precipitation events were also manifested in the soil moisture map time series. The machine learning method was quite robust since the pixelwise sensitivity to the choice of the method was estimated to be, on average, 6.3% with a standard deviation of 5.7% (in absolute soil moisture values).

Yet, the validity of the method could be checked only in the 92 different locations, for which soil moisture data was available: altogether 678 discrete values and eight points of continuous data. Hence, although the filtered S1 data used have a very high spatial resolution, ∼20 m, with a pixel size of 10 m, the spatially very limited *in situ* dataset does not allow us to conclude that the derived soil moisture maps would have the same soil moisture accuracy in the pixel resolution as the individual pixels used for developing and validating the soil moisture retrieval method.

To conclude, S1 has a high capability for observing seasonal soil moisture variation of subarctic forested and wetland areas. Furthermore, this methodology could be applied also to other subarctic and arctic regions to provide soil moisture estimates for the following purposes.

1) Ecohydrological and vegetation water usage studies.
2) To improve greenhouse gas estimates from forest lands.
3) To map influences of different land use actions such as logging at the landscape level.
4) To assess the forest fire risk.

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